

Pythagoras' Theorem & Trigonometry

Test · 27 marks total

Name: _____

Date: _____

Score: ____ / 27

Section A — Pythagoras' theorem

A1. A right-angled triangle has shorter sides of 5 cm and 12 cm. Find the length of the hypotenuse.

[2 marks]

A2. A right-angled triangle has a hypotenuse of 15 cm and one shorter side of 9 cm. Find the length of the other shorter side.

[3 marks]

A3. A rectangle measures 7 cm by 24 cm. Find the length of its diagonal.

[2 marks]

Section B — SOH CAH TOA basics

B1. In a right-angled triangle, which side is always the hypotenuse?

[1 mark]

B2. Write down the SOH CAH TOA formula you would use to find the opposite side if you know the hypotenuse and the angle.

[1 mark]

B3. Write down the SOH CAH TOA formula you would use to find an angle if you know the opposite and adjacent sides.

[1 mark]

Section C — Finding missing sides

C1. A right-angled triangle has an angle of 42° and a hypotenuse of 10 cm. Find the length of the side opposite the angle, to 1 decimal place.

[3 marks]

C2. A right-angled triangle has an angle of 55° and an adjacent side of 8 cm. Find the length of the hypotenuse, to 1 decimal place.

[2 marks]

C3. A right-angled triangle has an angle of 33° and an adjacent side of 6 cm. Find the length of the side opposite the angle, to 1 decimal place.

[2 marks]

Section D — Finding missing angles

D1. A right-angled triangle has an opposite side of 6 cm and a hypotenuse of 13 cm. Find the angle, to 1 decimal place.

[2 marks]

D2. A right-angled triangle has an adjacent side of 5 cm and an opposite side of 9 cm. Find the angle, to 1 decimal place.

[3 marks]

Section E — Applying Pythagoras & trigonometry

E1. A ladder rests against a vertical wall. The foot of the ladder is 1.8 m from the wall and the ladder is 4.5 m long. Calculate how far up the wall the ladder reaches, to 1 decimal place.

[2 marks]

E2. A flagpole casts a shadow. A wire from the top of the flagpole to a point on the ground 6 m from its base makes an angle of 58° with the ground. Calculate the height of the flagpole, to 1 decimal place.

[3 marks]